Lecture 6

Application I: point groups and molecular vibrations

Objectives. *The machinery of Lectures 3–5 now earns its keep on a concrete physical problem: the small oscillations of a molecule about equilibrium. We set up the dynamical matrix from first principles, let the point group act on displacements, and prove the central theorem of the subject: symmetry block-diagonalises the dynamical matrix, so that vibrational frequencies organise into families labelled by irreducible representations. Their dimensions give symmetry-enforced multiplet sizes; reduced-matrix multiplicity and the reality of the physical displacement space determine the total degeneracy. The programme is carried out in full for water and ammonia, including explicit symmetry-adapted modes from projection operators.*

6.1 Small oscillations and the dynamical matrix

Consider a molecule of N atoms with masses m_a and equilibrium positions $\vec{R}_a^{\,0}$, $a = 1, \ldots, N$, interacting through a potential V that depends only on the instantaneous positions $\vec{R}_a = \vec{R}_a^{\,0} + \vec{u}_a$ — and, the molecule being isolated, only through the interatomic distances, so that V is invariant under rigid Euclidean motions of the whole molecule (this is what Section 6.3 uses). Equilibrium means the gradient vanishes there, so for small displacements Taylor's theorem gives

$$V = V_0 + \frac{1}{2} \sum_{a\alpha,\, b\beta} \frac{\partial^2 V}{\partial u_{a\alpha}\, \partial u_{b\beta}}\bigg|_0 u_{a\alpha}\, u_{b\beta} + O(u^3),$$

with $\alpha, \beta \in \{x, y, z\}$ Cartesian indices: to leading order, every molecule is a system of coupled harmonic oscillators. Newton's equations $m_a \ddot{u}_{a\alpha} = -\partial V / \partial u_{a\alpha}$ carry awkward masses; they disappear in the *mass-weighted coordinates* $q_{a\alpha} = \sqrt{m_a}\, u_{a\alpha}$, in terms of which

$$\ddot{q} = -D\, q, \qquad D_{a\alpha,\, b\beta} = \frac{1}{\sqrt{m_a m_b}} \frac{\partial^2 V}{\partial u_{a\alpha}\, \partial u_{b\beta}}\bigg|_0 . \tag{6.1}$$

The $3N \times 3N$ real symmetric matrix D is the *dynamical matrix*. (The symbol D is entrenched in physics for this object, as $D(g)$, $D^{(i)}(g)$ are for the representation matrices of Lectures 3–5; the two will never meet in one formula, and an argument g or a superscript always signals the latter.)

Being real symmetric, D has an orthonormal eigenbasis, $D\, \varepsilon^{(k)} = \omega_k^2\, \varepsilon^{(k)}$, and the general solution of (6.1) is a superposition of *normal modes*

$$q(t) = \varepsilon^{(k)} \cos\big(\omega_k t + \varphi_k\big) :$$

collective motions in which every atom oscillates with one common frequency ω_k and fixed amplitude pattern $\varepsilon^{(k)}$. At a stable equilibrium $D \geq 0$, so all $\omega_k^2 \geq 0$; the rigid motions of the molecule will turn out to be modes with $\omega_k = 0$ (Section 6.3), and we assume the equilibrium is stable in the strict sense that these are the *only* zero modes (D positive definite transverse to them); a floppy molecule with further zero modes needs separate treatment. The questions this lecture answers by pure symmetry: how do the $3N$ modes organise, which frequencies are forced to coincide, and what do the patterns $\varepsilon^{(k)}$ look like — all *before* any spring constant is known.

6.2 The displacement representation

The configuration of small displacements is the vector $u = (\vec{u}_1, \ldots, \vec{u}_N)$, an element of the $3N$-dimensional real *displacement space* $V_{\mathrm{disp},\mathbb{R}}$. We apply the complex character theory of Lectures 3–5 to its complexification $V_{\mathrm{disp}} = V_{\mathrm{disp},\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$, the space of vectors $u + \mathrm{i}v$ with u, v real. Complex conjugation $C(u + \mathrm{i}v) = u - \mathrm{i}v$ on this space fixes the physical real vectors, commutes with the real point-group matrices and with the real dynamical matrix, and will therefore relate conjugate complex irreducibles. The point group G of the equilibrium configuration acts on this space, and the action is an instance — the course's third — of the function-space construction of Lecture 3. A displacement field is a vector-valued function on the set of atoms, $a \mapsto \vec{u}_a$; a symmetry operation $g \in G \subset \mathrm{O}(3)$, with 3×3 matrix $R(g)$, moves atom a to the site of atom $g(a)$ (a permutation of identical nuclei) and rotates displacement vectors. The transported field assigns to the *new* site the rotated displacement of the *old*:

$$\big(\rho(g)\, u\big)_a = R(g)\, \vec{u}_{g^{-1}(a)}, \tag{6.2}$$

the familiar inverse on the argument, with the new feature that the value is rotated too. The homomorphism property follows exactly as in Lecture 3. Since g maps each atom to one of equal mass, the matrices of $\rho(g)$ are the same in mass-weighted as in plain coordinates, and they are orthogonal (a permutation of blocks, each block the orthogonal $R(g)$).

Everything we need from this representation is its character, and the fixed-point lemma of Lecture 5 delivers it instantly, upgraded by the 3×3 trace:

Lemma 6.1 (Atoms left in place)**.** *Let N_g be the number of atoms left in place by the operation g (atoms fixed by the full operation: on its axis for a rotation, in its plane for a reflection, at the centre for the inversion or for a rotoreflection $S(\theta)$ with $\theta \neq 0$). Then*

$$\chi_{\mathrm{disp}}(g) = N_g \cdot \chi_{\mathrm{vec}}(g), \quad \textit{where} \quad \chi_{\mathrm{vec}}(g) = \operatorname{tr} R(g) = \begin{cases} 1 + 2\cos\theta, & g = C(\theta), \\ -1 + 2\cos\theta, & g = S(\theta), \end{cases}$$

with $C(\theta)$ a proper rotation through θ and $S(\theta)$ an improper operation (rotoreflection).

Proof. In the basis labelled by (atom, direction), the matrix of $\rho(g)$ has 3×3 blocks: by (6.2), block row a, block column b contains $R(g)\, \delta_{a, g(b)}$. Diagonal blocks occur exactly for atoms with $g(a) = a$, each contributing $\operatorname{tr} R(g)$; moved atoms contribute nothing to the trace. For the 3×3 traces: a proper rotation through θ has, in a basis adapted to its axis, the matrix $\mathrm{diag}\big(R_{2\times 2}(\theta), 1\big)$ with trace $1 + 2\cos\theta$; an improper operation is, by Lecture 2's factorisation $\mathrm{O}(3) = \mathrm{SO}(3) \times \{\pm\mathbb{1}\}$, of the form $-\mathbb{1} \cdot C$ with C a rotation — more directly, every improper operation is a *rotoreflection* $S(\theta)$, rotation through θ composed with reflection in the perpendicular plane, $\mathrm{diag}\big(R_{2\times 2}(\theta), -1\big)$, of trace $2\cos\theta - 1$. The trace is basis-independent, so the adapted basis costs nothing. ■

The values needed in practice:

g	E	C_2	C_3	C_4	C_6	$\sigma = S(0)$	$i = S(\pi)$
$\chi_{\mathrm{vec}}(g)$	3	−1	0	1	2	1	−3

6.3 Discarding rigid motions

Six of the $3N$ degrees of freedom are not vibrations. A rigid translation, $\vec{u}_a = \vec{t}$ for all a, and an infinitesimal rigid rotation, $\vec{u}_a = \delta\vec{\omega} \times \vec{R}_a^0$, change no interatomic distance, hence leave V *exactly* invariant: they are zero modes of D, with $\omega = 0$, and they span invariant subspaces of the displacement representation (a symmetry operation maps rigid motions to rigid motions). The genuine vibrations live in the complementary invariant subspace, and at the level of characters the separation is a subtraction:

$$\chi_{\mathrm{vib}} = \chi_{\mathrm{disp}} - \chi_{\mathrm{trans}} - \chi_{\mathrm{rot}}.$$

The two subtracted characters are easy. Translations are labelled by the vector $\vec{t}$, which transforms by $R(g)$ itself: $\chi_{\mathrm{trans}} = \chi_{\mathrm{vec}}$. Rotations are labelled by $\delta\vec{\omega}$, which is an *axial* vector, and axial vectors acquire a determinant:

Lemma 6.2 (Axial vectors). *For $R \in \mathrm{O}(3)$, $(R\vec{a}) \times (R\vec{b}) = \det(R)\ R\,(\vec{a} \times \vec{b})$. Consequently $\delta\vec{\omega} \mapsto \det\big(R(g)\big)R(g)\,\delta\vec{\omega}$, and*

$$\chi_{\mathrm{rot}}(g) = \det\big(R(g)\big)\,\chi_{\mathrm{vec}}(g) = \begin{cases} 1 + 2\cos\theta, & g \text{ proper}, \\ 1 - 2\cos\theta, & g \text{ improper}. \end{cases}$$

Proof. For proper R the identity is geometry: rotations preserve lengths, angles and orientation, and the cross product is defined by exactly those data, so it commutes with rotations. A general orthogonal matrix is R or $-R$ with R proper (Lecture 2), and $(-\vec{a}) \times (-\vec{b}) = \vec{a} \times \vec{b}$ supplies the factor $\det(-\mathbb{1}) = -1$ relative to the polar transformation law. Writing $\vec{u}_a = \delta\vec{\omega} \times \vec{R}_a^0$ and transforming both the displacement (polar) and the sites, consistency forces $\delta\vec{\omega}$ to transform with the extra det. ∎

For a *nonlinear* molecule the rotation labels $\delta\vec{\omega}$ fill three dimensions and

$$\dim V_{\mathrm{vib}} = 3N - 6;$$

a linear molecule has only two independent rotations (spinning about its own axis moves nothing) and $3N - 5$ vibrations — a caveat that matters for CO_2.

6.4 The central theorem: symmetry block-diagonalises D

That G is a symmetry *of the molecule* — it permutes identical nuclei and preserves all interatomic geometry — means precisely that the potential is invariant: $V\big(\rho(g)u\big) = V(u)$ for all $g \in G$ and all displacements. For the quadratic form $V = \frac{1}{2}\,q^{\mathsf{T}} D q$ this reads $\rho(g)^{\mathsf{T}} D\,\rho(g) = D$, and since the matrices $\rho(g)$ are orthogonal,

$$\big[D,\ \rho(g)\big] = 0 \qquad \text{for all } g \in G. \tag{6.3}$$

The dynamical matrix is an intertwiner of the displacement representation with itself — and Lectures 4–5 have told us exactly how much freedom such an operator has.

Theorem 6.3 (Symmetry structure of the vibrational spectrum)**.** *Decompose the vibrational representation into irreducibles, $V_{\mathrm{vib}} \cong \bigoplus_i m_i V^{(i)}$, and choose in each isotypic component V_i a symmetry-adapted orthonormal basis: vectors $w^{(a)}_\mu$, $a = 1, \ldots, m_i$, $\mu = 1, \ldots, n_i$, each copy transforming with the same unitary matrices, $\rho(g)\, w^{(a)}_\nu = \sum_\mu D^{(i)}_{\mu\nu}(g)\, w^{(a)}_\mu$, and $\left\langle w^{(a)}_\mu, w^{(b)}_\nu \right\rangle = \delta_{ab}\delta_{\mu\nu}$ (such a basis exists: split V_i orthogonally into m_i copies using the invariant inner product, and transport an orthonormal adapted basis of one copy to the others by intertwiners, which are unitary up to scale by Schur's lemma). Then:*

(i) D preserves every isotypic component, and has no matrix elements between different components or different partner indices:

$$\left\langle w^{(a)}_\mu,\, D\, w^{(b)}_\nu \right\rangle = F^{(i)}_{ab}\, \delta_{\mu\nu},$$

with an $m_i \times m_i$ Hermitian reduced matrix *$F^{(i)}$ that is independent of the partner index μ;*

(ii) if an eigenvalue ω^2 of $F^{(i)}$ has multiplicity q, then its total complex degeneracy in this isotypic component is qn_i; the factor $n_i = \dim_{\mathbb{C}} V^{(i)}$ is the symmetry-enforced multiplet size;

(iii) if every reduced eigenvalue is simple and eigenvalues from unrelated reduced blocks do not coincide, the complex multiplets have size n_i. Repeated reduced eigenvalues contribute the factor q, while coincidences between unrelated blocks are accidental. For a real physical system the conjugate-pair constraints below may enforce additional pairing.

Proof. D commutes with each projector P_i, which is a linear combination of the $\rho(g)$ (Lecture 5); hence $DP_i = P_iD$ and D maps $V_i = \operatorname{im} P_i$ to itself, with $P_jDP_i = DP_jP_i = 0$ for $j \neq i$: no cross-component matrix elements. Within V_i, expand

$$D\, w^{(b)}_\nu = \sum_{a,\,\mu} K^{ab}_{\mu\nu}\, w^{(a)}_\mu .$$

Apply $\rho(g)$ to both sides and use (6.3) together with the adapted transformation law; comparing coefficients of $w^{(a)}_\mu$ gives, for every g and every pair (a, b),

$$\sum_\lambda K^{ab}_{\mu\lambda}\, D^{(i)}_{\lambda\nu}(g) = \sum_\lambda D^{(i)}_{\mu\lambda}(g)\, K^{ab}_{\lambda\nu} :$$

each $n_i \times n_i$ matrix K^{ab} commutes with the irreducible representation $D^{(i)}$, hence by Schur's lemma $K^{ab} = F_{ab}\, \mathbb{1}$, i.e. $K^{ab}_{\mu\nu} = F_{ab}\, \delta_{\mu\nu}$. This is (i); Hermiticity of $F^{(i)}$ is inherited from D.

For (ii), diagonalise $F^{(i)}$: for each vector c in a q-dimensional ω^2-eigenspace, the vectors $\sum_b c_b\, w^{(b)}_\mu$ for $\mu = 1, \ldots, n_i$ are n_i linearly independent eigenvectors of D with the common eigenvalue ω^2; varying c gives qn_i independent eigenvectors — one for each partner index, all with the *same* reduced eigenvalue because $F^{(i)}$ did not depend on μ. Conversely every eigenvector of D in V_i arises this way, since the $w^{(a)}_\mu$ exhaust V_i. Part (iii) separates the generic simplicity hypothesis from the two possible kinds of coincidence. ∎

Remark 6.4 (Reality constraints and complex irreducibles)**.** The theorem is complex-linear, but the molecule is real. Let C denote complex conjugation on the complexified displacement space. Since $CD = DC$ and $C\rho(g) = \rho(g)C$, conjugation carries the isotypic component of $V^{(i)}$ to that of its conjugate $\overline{V^{(i)}}$. In adapted bases of the two components related by C (*conjugate adapted bases*) the reduced blocks obey $F^{(\bar{i})} = \overline{F^{(i)}}$, so non-real conjugate irreducibles have the same spectrum and combine into real multiplets. A complex one-dimensional sector therefore need not give a nondegenerate physical mode.

In the brief real-type language (Frobenius–Schur; stated without proof): an irrep is of *real type* if it is equivalent to one by real matrices, of *complex type* if it is not equivalent to its conjugate, and of *quaternionic type* if it is self-conjugate but has no real form — the three cases being told apart by $\frac{1}{|G|}\sum_g \chi(g^2) = 1, 0, -1$. A self-conjugate complex irrep of real type has a real form; a non-self-conjugate irrep is of complex type and must travel with its conjugate; a self-conjugate irrep of quaternionic type has no real form and occurs with the corresponding even-multiplicity constraint when a real representation is complexified. We keep the complex character-table calculus, adding only these reality constraints.

The planar rotation representation of C_3 is the diagnostic example. On $\mathbb{R}^2$, let c act by $R(2\pi/3)$ and $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. After complexification the representation is the sum of the conjugate one-dimensional characters $\chi_1 \oplus \chi_2$. A real matrix commuting with the rotation has the form $a\mathbb{1} + bJ$; if it is also symmetric, as a dynamical matrix is, then $b = 0$. Thus $D = a\mathbb{1}$: the two conjugate complex singlets form a reality-enforced real doublet, not an accidental coincidence.

Three comments. First, the division of labour: *group theory fixes the block structure — the sizes m_i, the forced multiplet factors n_i, the symmetry-adapted bases — while the dynamics (spring constants, masses) only fills in the numerical entries of the small reduced matrices $F^{(i)}$*. A $3N$-dimensional eigenvalue problem collapses into independent $m_i \times m_i$ problems. Second, a degeneracy larger than n_i can come from a repeated eigenvalue inside one $F^{(i)}$, from an accidental coincidence between unrelated blocks, or from the reality pairing of Remark 6.4; only the last is an additional physical constraint. Lecture 7 takes this theme into quantum mechanics. Third, the mechanism of the proof — an equivariant operator forced by Schur's lemma to be scalar across partner indices, leaving one reduced matrix element per pair of copies — is worth committing to memory: it is the Wigner–Eckart theorem in embryo.

The practical recipe, then:

(1) identify the point group G of the equilibrium configuration;
(2) compute $\chi_{\mathrm{disp}}(g) = N_g\,\chi_{\mathrm{vec}}(g)$ class by class;
(3) subtract χ_{trans} and χ_{rot};
(4) extract multiplicities $m_i = \langle \chi_i, \chi_{\mathrm{vib}} \rangle_G$ from the character table;
(5) diagonalise each reduced $F^{(i)}$: a reduced eigenvalue of multiplicity q gives complex degeneracy qn_i, then impose the conjugate-pair reality constraints;
(6) construct explicit modes with the projectors of Lecture 5.

6.5 Worked example: water

Water is bent: $N = 3$, point group $C_{2v} = \{E, C_2, \sigma_v, \sigma_v'\}$ (Lecture 1). We place the molecule in the yz-plane with the C_2 axis along z; σ_v denotes the molecular plane (yz) and σ_v' the perpendicular plane (xz). The group is abelian of order 4, so its four irreducible characters are one-dimensional sign characters; with Mulliken labels:

C_{2v}	E	C_2	$\sigma_v(yz)$	$\sigma'_v(xz)$	
A_1	1	1	1	1	z
A_2	1	1	−1	−1	
B_1	1	−1	1	−1	y
B_2	1	−1	−1	1	x

(the last column records how the coordinate functions transform, needed shortly; beware that interchanging the two planes interchanges the labels $B_1 \leftrightarrow B_2$ — conventions differ between books, so fix one and stay consistent).

Step 1: characters. The operation counts and traces:

	E	C_2	$\sigma_v(yz)$	$\sigma'_v(xz)$
N_g	3	1	3	1
χ_{vec}	3	−1	1	1
$\chi_{\text{disp}} = N_g\chi_{\text{vec}}$	9	−1	3	1
χ_{trans}	3	−1	1	1
$\chi_{\text{rot}} = \det\cdot\chi_{\text{vec}}$	3	−1	−1	−1
χ_{vib}	3	1	3	1

($N_{C_2} = 1$: only the oxygen sits on the axis; $N_{\sigma_v} = 3$: the whole molecule lies in its own plane; $N_{\sigma'_v} = 1$: only the oxygen lies in the perpendicular plane.)

Step 2: multiplicities. With all class sizes 1 and $|G| = 4$:

$$m_{A_1} = \tfrac{1}{4}(3+1+3+1) = 2, \qquad m_{A_2} = \tfrac{1}{4}(3+1-3-1) = 0,$$
$$m_{B_1} = \tfrac{1}{4}(3-1+3-1) = 1, \qquad m_{B_2} = \tfrac{1}{4}(3-1-3+1) = 0,$$

so

$$\Gamma_{\text{vib}} = 2A_1 \oplus B_1,$$

three vibrational coordinates ($3N-6 = 3$ ✓): *two* of full symmetry A_1 and one antisymmetric under both C_2 and σ'_v. Generically the 2×2 A_1 reduced block has two distinct eigenvalues, so all three frequencies are nondegenerate; an equal pair inside that block would be an accidental reduced-eigenvalue coincidence.

Step 3: explicit modes by projection. The vibrational physics is most transparent in internal coordinates (linear functions of the displacements that parametrise V_{vib} and that G permutes; they carry an equivalent copy of the vibrational representation): the two bond stretches s_1, s_2 (elongations of the O–H bonds) and the bend δ (the H–O–H angle increment) — three coordinates for three vibrations. The group permutes them simply: C_2 and σ'_v swap $s_1 \leftrightarrow s_2$, while E and σ_v fix each; δ is invariant under everything (hence pure A_1). The Lecture-5 projectors applied to the trial vector s_1:

$$P_{A_1}s_1 = \tfrac{1}{4}\big(s_1+s_2+s_1+s_2\big) = \tfrac{1}{2}(s_1+s_2),$$
$$P_{B_1}s_1 = \tfrac{1}{4}\big(s_1+(-1)s_2+s_1+(-1)s_2\big) = \tfrac{1}{2}(s_1-s_2),$$

with $P_{A_2}s_1 = P_{B_2}s_1 = 0$. The three symmetry-adapted vibrations of water (Figure 6.1):

$$\underbrace{s_1+s_2}_{\text{symmetric stretch }(A_1)}, \qquad \underbrace{\delta}_{\text{bend }(A_1)}, \qquad \underbrace{s_1-s_2}_{\text{antisymmetric stretch }(B_1)}.$$

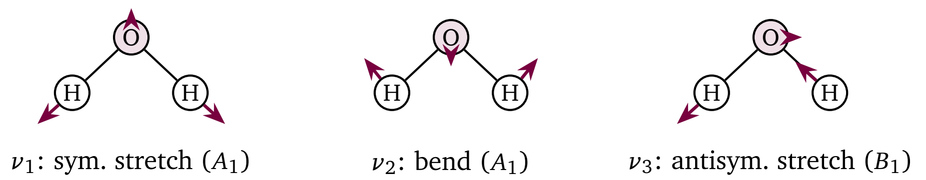

Figure 6.1: The three normal modes of water with their Mulliken labels. Small arrows on the oxygen keep the centre of mass fixed. The two A_1 coordinates mix; the patterns shown are the symmetry-adapted coordinates whose 2×2 reduced matrix determines the two A_1 frequencies.

Theorem 6.3 adds a final refinement that pure mode pictures gloss over: the two A_1 coordinates span a single isotypic component with $m_{A_1} = 2$, so the symmetric stretch and the bend are *not* individually normal modes — the actual eigenvectors are the two orthogonal combinations determined by the 2×2 reduced matrix $F^{(A_1)}$, whose entries depend on the force constants. The B_1 mode, alone in its symmetry type, is an exact normal mode by symmetry alone. Group theory has reduced a 9×9 problem to a 2×2 problem (A_1), a 1×1 problem (B_1), and the six rigid modes already known to have zero frequency.

6.6 Worked example: ammonia

Ammonia is a triangular pyramid: $N = 4$, point group C_{3v} with the character table of Lecture 5,

C_{3v}	E	$2C_3$	$3\sigma_v$	
A_1	1	1	1	z
A_2	1	1	-1	
E	2	-1	0	(x, y)

(z along the threefold axis). The character bookkeeping, with class sizes $1, 2, 3$:

	E	$2C_3$	$3\sigma_v$
N_g	4	1	2
χ_{vec}	3	0	1
χ_{disp}	12	0	2
χ_{trans}	3	0	1
χ_{rot}	3	0	-1
χ_{vib}	6	0	2

($N_{C_3} = 1$: nitrogen on the axis; $N_{\sigma_v} = 2$: nitrogen plus the one hydrogen contained in each vertical plane). Multiplicities, with $|G| = 6$:

$$
\begin{aligned}
m_{A_1} &= \tfrac{1}{6}\big[1 \cdot 1 \cdot 6 + 2 \cdot 1 \cdot 0 + 3 \cdot 1 \cdot 2\big] = 2,\\
m_{A_2} &= \tfrac{1}{6}\big[6 + 0 - 6\big] = 0,\\
m_E &= \tfrac{1}{6}\big[1 \cdot 2 \cdot 6 + 2 \cdot (-1) \cdot 0 + 3 \cdot 0 \cdot 2\big] = 2,
\end{aligned}
$$

so

$$\Gamma_{\text{vib}} = 2A_1 \oplus 2E :$$

six vibrational dimensions ($3N - 6 = 6$ ✓) and generically four distinct frequencies — two nondegenerate A_1 frequencies and two doubly degenerate E frequencies. This is exactly the observed infrared spectrum of ammonia: the symmetric stretch and the umbrella mode (A_1), and the doubly degenerate asymmetric stretch and asymmetric bend (E). The double degeneracies are not numerical accidents to be checked against force fields; they are theorems.

For explicit modes, internal coordinates again serve best: the three bond stretches s_1, s_2, s_3 and the three inter-bond angle increments $\delta_{12}, \delta_{23}, \delta_{31}$ — six coordinates for six vibrations. Under $C_{3v} \cong S_3$ each triple is simply permuted: the rotations cycle the indices, the reflection through H_1 swaps $2 \leftrightarrow 3$. But the permutation representation of S_3 on three objects is an old friend: its character is the fixed-point count $(3, 0, 1)$, and by the one-line computation of Lecture 5 (or by inspection of the C_{3v} table above) it decomposes as

$$(3, 0, 1) = A_1 \oplus E.$$

Each triple therefore contributes one A_1 and one E, giving $2A_1 \oplus 2E$ in agreement with the Cartesian count — and the projectors are *verbatim* those computed for the S_3 permutation representation in Lecture 5, $P_{A_1} = J/3$ and $P_E = \mathbb{1} - J/3$ on each triple:

$$\underbrace{s_1 + s_2 + s_3}_{\text{symmetric stretch } (A_1)}, \qquad \underbrace{\delta_{12} + \delta_{23} + \delta_{31}}_{\text{umbrella } (A_1)},$$

$$\underbrace{\big(2s_1 - s_2 - s_3,\ s_2 - s_3\big)}_{\text{asymmetric stretch, } E \text{ partners}}, \qquad \underbrace{\big(2\delta_{23} - \delta_{31} - \delta_{12},\ \delta_{31} - \delta_{12}\big)}_{\text{asymmetric bend, } E \text{ partners}}.$$

As with water, the two A_1 coordinates mix through a 2×2 reduced matrix, as do the two E pairs (partner by partner, with the *same* 2×2 matrix for both partners — that sameness is the content of Theorem 6.3(i) and the origin of the symmetry-enforced doublet factor). If the two eigenvalues of this reduced block happened to coincide, the total E degeneracy would be four. A 12×12 eigenvalue problem has become two 2×2 problems.

Remark 6.5 (Internal coordinates: a caveat)**.** For water and ammonia the natural internal coordinates number exactly $3N - 6$ and the bookkeeping closes. For other molecules (rings, high-coordination centres) natural internal coordinates can be *redundant*; the character calculus detects this automatically — the internal-coordinate representation then properly contains Γ_{vib}, and the surplus must be identified and discarded. The Cartesian route of Sections 6.2–6.3 is never ambiguous.

6.7 Infrared and Raman activity: a preview

Which vibrations does a spectrometer actually see? Light couples to the molecular dipole moment, whose components transform like the coordinate functions (x, y, z); light scattering (Raman) couples to the polarisability, whose components transform like the quadratics $(x^2, y^2, z^2, xy, yz, zx)$. The full justification is the selection-rule machinery of Lecture 7, but the rule itself can be stated now:

> *A vibrational mode is infrared active iff its irreducible representation occurs among those of (x, y, z), and Raman active iff it occurs among those of the quadratics.*

For water, the table in Section 6.5 shows (x, y, z) spanning $B_2 \oplus B_1 \oplus A_1$; all three vibrations ($2A_1 \oplus B_1$) match and are infrared active — as observed, water absorbs at all three fundamentals (the reason it is a greenhouse gas). For ammonia, (x, y, z) spans $A_1 \oplus E$, so all four fundamentals $2A_1 \oplus 2E$ are infrared active — also as observed. The rule has teeth in molecules with more symmetry: in centrosymmetric molecules (point group containing the inversion i) the coordinates are odd under i while the quadratics are even, so *no mode is both infrared and Raman active* — the mutual exclusion rule, which Lecture 7 will derive in two lines from parity and which Sheet 6 explores.

Remark 6.6 (Computer algebra: numerics against theorems)**.** The accompanying notebook closes the loop between group theory and dynamics. It models water and ammonia with harmonic bond and angle springs, builds the Hessian by symbolic differentiation, mass-weights it, and calls `Eigensystem`. The numerical spectra display the predicted generic shapes — three distinct frequencies for water; four frequencies for ammonia of which two are double to machine precision — for generic spring constants. Tuned parameters can repeat a reduced eigenvalue or align unrelated blocks. Applying the projector matrices P_i of Lecture 5 (built from the displacement representation (6.2)) to the numerical eigenvectors confirms each one's symmetry label, and conjugating D into a symmetry-adapted basis displays the block structure of Theorem 6.3: the 9×9 and 12×12 dynamical matrices fall apart into the predicted small blocks, with the E blocks of ammonia appearing in identical pairs.

Summary.

- *Vibrations as a representation.* Nuclear displacements carry a (generally reducible) representation of the molecule's point group (eq. (6.2)).
- *Symmetry commutes.* The point group commutes with the dynamical matrix (eq. (6.3)), so each frequency eigenspace is a representation.
- *Modes by irreducible.* The complex irrep dimension is the symmetry-enforced multiplet factor; a reduced eigenvalue of multiplicity q has total complex degeneracy $q \dim V^{(i)}$, with conjugate sectors paired by reality (Theorem 6.3, Remark 6.4).
- *Mode counting.* The displacement character is read off the atoms a symmetry leaves in place (Lemma 6.1); removing the translations and rotations (Lemma 6.2) leaves the vibrational content.
- *Symmetry-adapted modes.* Character projectors construct the normal-mode coordinates, as worked for H_2O and NH_3.

Tutorial. Exercise Sheet 6.

References

Bishop, *Group Theory and Chemistry*; Tinkham, *Group Theory and Quantum Mechanics* (molecular vibrations); Elliott–Dawber, *Symmetry in Physics* (vibrational analysis).

Bibliography

The per-lecture *References* sections cite the works below by author and short title; full details are collected here. Chapter and section numbers in the lecture references follow these editions.

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- A. O. Barut and R. Rączka, *Theory of Group Representations and Applications*, 2nd ed. World Scientific, 1986.
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